Question:

A 9-month-old boy is brought to the pediatrician because he can not sit on his own without support and has involuntary movements. He was born vaginally with no complications at full term. There is no history of consanguinity among parents. On physical examination, it was noticed that he is a stunted infant with generalized hypotonia and severe generalized dystonic movements. The mother says that she has noticed the presence of orange sand in his diapers many times. Laboratory evaluation revealed elevated uric acid levels in both blood and urine. Hypoxanthine-guanine phosphoribosyltransferase is found to be deficient in his blood samples. He was prescribed an appropriate medication and sent home. The most likely mechanism of this drug is the inhibition of which of the following enzymes in addition to xanthine oxidase?Answer Choices:A. Purine nucleoside phosphorylaseB. Orotate phosphoribosyltransferaseC. Ribonucleotide reductaseD. Dihydrofolate reductase

Reasoning + Answer:

<think>thecompoundsinthepurinepathwayhavebeenidentifiedashypoxanthine , xanthine ,
andthenuricacid . henceifuricacidiselevateditmeansthatpathwayisblocked . uricacidis
alsomadebyhypoxanthine-guaninephosphoribosyltransferasesothesecondreactionislikely
inhibitedinadditiontoxanthineoxidase . thecompoundnvolvedinthesecondreactionisguanine
. </think><answer>Guaninephosphoribosyltransferase</answer>